EXPERIMENT 9:

PROGRAM CODE:

clc; clear; close all;

% Parameters fs = 100000; t = 0:1/fs:0.02; fm = 500; fc = 5000;

% Message Signal m = cos(2*pi*fm*t);

% Carrier Signal c = cos(2*pi*fc*t);

% AM Modulation am = (1 + m).*c;

% Add AWGN Noise (10 dB SNR) noisy = am + randn(size(am))*sqrt(mean(am.^2)/10);

% Envelope Detection env = abs(hilbert(noisy));

% Display Results figure;

subplot(5,1,1); plot(t,m); title('Message Signal'); grid on;

subplot(5,1,2); plot(t,c); title('Carrier Signal'); grid on;

subplot(5,1,3); plot(t,am); title('AM Signal'); grid on;

subplot(5,1,4); plot(t,noisy); title('AM Signal with AWGN Noise'); grid on;

subplot(5,1,5); plot(t,env); title('Recovered Signal'); grid on;

xlabel('Time (seconds)');

OUTPUT :

